David Hu

571-551-9994 | who.is.david101@gmail.com | linkedin.com/in/davidhu426 | github.com/dhu2022-dev

Education

University of Virginia Charlottesville, VA
Bachelor's in Computer Science *Aug. 2022 – Aug. 2026*

- Notable Electives: Advanced Algorithms and Implementations, Machine Learning, Artificial Intelligence, Cloud Computing, Databases, Software Testing

Experience

Software Engineer Contractor Mar. 2026 – May 2026
Mercor *Remote*

- Built internal Python tooling for multi-agent AI experimentation, implementing scenario orchestration, observer-based evaluation pipelines, live transcript visualization, and experiment logging for coordination-failure analysis across simulated agent interactions.
- Contributed to multimodal AI evaluation pipelines for frontier video understanding models by designing and quality-calibrating RL training data across 5 reasoning domains, supporting a 4,000-sample dataset initiative with planned scaling to 10,000+ video QA samples.
- Developed structured QA standards, annotation taxonomies, and ambiguity-reduction workflows for video reasoning datasets spanning temporal localization, counting, causal reasoning, action recognition, and audio-visual grounding tasks.

Software Engineering Intern June 2025 – Aug. 2025
Capital One – OptiCloud Team, Cloud Platforms *Richmond, VA*

- Developed and deployed a production-grade anomaly detection system that identified faulty cloud optimization recommendations, boosting AWS spend efficiency by 35% and saving $20,000+ annually.
- Architected scalable ML pipelines using Databricks and AWS Glue, processing 10M+ training records across S3, PostgreSQL, and DynamoDB; deployed STL and Autoencoder models with SHAP-based explanations.
- Built and integrated Go services powering real-time health dashboards for FinOps and Cloud Governance teams, using config-driven APIs and dynamic visualization components.

AI Developer Intern May 2024 – Aug. 2024
American Chemical Society *Washington, D.C.*

- Developed an ML pipeline for journal classification using LDA topic modeling, BERTopic, and visualization workflows to support manuscript categorization and topic discovery.
- Created Python scripts to extract research funder and grant information from publications using spaCy, regex, and LLM-assisted workflows, reducing manual data entry time by 40%.
- Automated API data extraction and processing workflows using Docker, SQL, PostgreSQL, Microsoft Azure, and Doccano, improving article search relevance and reducing researcher data retrieval time.

Projects

UVAHacks Options Automation Tool – 1st Place | *Python, Streamlit, OAuth 2.0* April 2024

- Won Best Finance Hack at UVA's HooHacks 2024 for a real-time options analytics platform built for student investment workflows.
- Built a Streamlit web app for options analysis, portfolio visualization, and trade evaluation, integrating brokerage-style authentication and API-based market data retrieval.
- Implemented Black-Scholes and Fourier-based pricing models to support real-time options pricing, scenario analysis, and interactive investment decision-making.

HackMIT AI Notetaker | *Python, TensorFlow, OpenCV, Modal* September 2023

- Developed an AI-powered application at HackMIT that converted handwritten notes and lecture recordings into structured notes, summaries, and diagrams.
- Implemented OCR workflows using TensorFlow and OpenCV to extract text from note images and transform unstructured inputs into usable study materials.
- Integrated Modal-hosted image generation workflows to create diagrams from user prompts, earning an opportunity to pitch the project to Y Combinator representatives.

Skills and Certifications

Languages: Python, Java, Go, SQL, TypeScript
Frameworks and Tools: AWS, Databricks, AWS Glue, Docker, PostgreSQL, Django, REST APIs
ML/AI: Scikit-learn, BERTopic, LDA, TensorFlow, OpenCV, SHAP, Pandas
Other: Portrait Photography Patent (No. 11086197), ICPC Regionals Honorable Mention